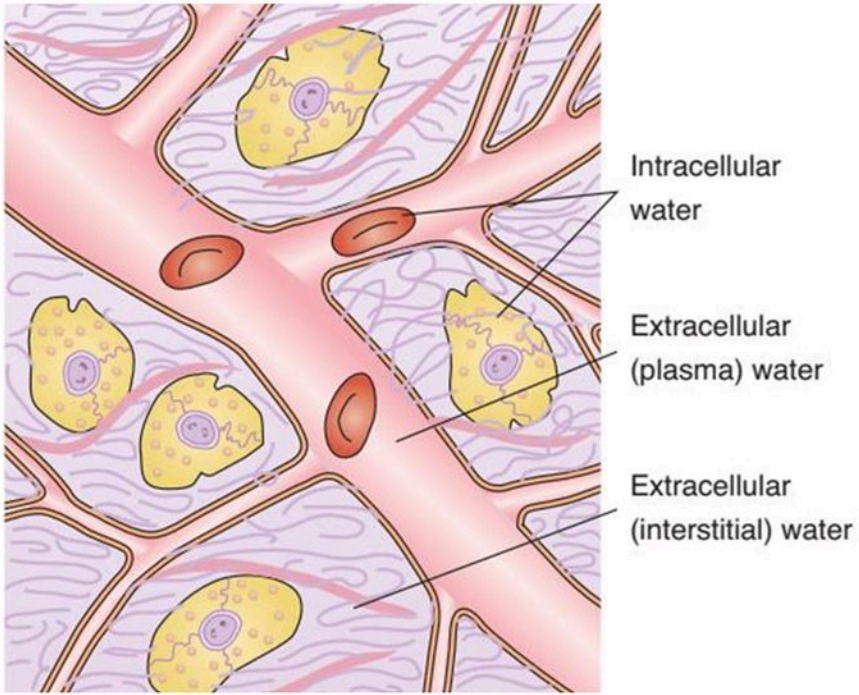
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Body Fluid Composition



Body fluid composition

- Body fluid composition is essential for understanding how fluids are distributed and regulated in the body. Here's a breakdown:
- **Main Fluid Compartments**:
- **Intracellular Fluid (ICF)**: Comprises about 40% of body weight.
- **Extracellular Fluid (ECF)**: Divided into:
  - **Intravascular Fluid** (plasma): Accounts for about 5% of body weight.
  - **Interstitial Fluid**: Makes up around 25% of body weight.

Body fluid composition

- **Water Proportion by Age**:
- In a term newborn, water constitutes about 78% of body weight.
- This decreases to approximately 60% in adults.
- **Water Balance**:
- Depends on intake, output, and metabolic usage.
- Water is typically lost via:
  - Skin
  - Lungs
  - Intestine
  - Kidneys

Body fluid composition

- **Cell Membrane Permeability**:
- Cell membranes are relatively permeable to **water**, **potassium (K^+)**, and **chloride (Cl^-)**.
- They are relatively impermeable to **sodium (Na^+)**, **phosphate (PO_4^{3-})**, and **proteins**.
- Sodium (Na^+) is actively transported out of cells via sodium pumps, which require energy.

Body fluid composition

- **Fluid Movement and Equilibrium**:
- Tonicity, hydrostatic pressure, and colloid osmotic pressure between plasma and interstitial fluid help maintain fluid balance.
- At the **arterial end of capillaries**, water exits under hydrostatic pressure.
- At the **venous end**, water is drawn back in by plasma oncotic pressure.

Body fluid composition

- **Plasma Osmolality**:
- Plasma osmolality remains fairly constant, typically between **285 and 300 mOsmol/kg H_2O**, helping maintain cellular homeostasis and fluid balance.

🔴 **High Plasma Osmolality (> 300 mOsm/kg H_2O) – Dehydration**

- **Causes:** Excessive sweating, vomiting, diarrhea, uncontrolled diabetes.
- **Effects:** Cells shrink due to water loss, leading to **confusion, weakness, and possible coma**.

🔵 **Low Plasma Osmolality (< 285 mOsm/kg H_2O) – Overhydration**

- **Causes:** Excessive water intake, kidney disease, SIADH (syndrome of inappropriate ADH secretion).
- **Effects:** Cells swell, which can lead to **brain swelling (cerebral edema), headaches, seizures, and coma**.

Calculating Pediatric Fluid Needs

- Calculating pediatric fluid needs is essential for ensuring adequate hydration, especially for children in clinical settings. Here's a summary of the maintenance fluid calculation method based on weight, as adapted from South & Isaacs (2012):

Pediatric Maintenance Fluid Calculation

- **For children under 10 kg:**
- Fluid requirement: **100 ml per kilogram of body weight per day**
- Formula: Fluid needs=Weight in kg×100 ml/kg/day
- **Example**: For a child weighing 7 kg:
  7 kg×100 ml=700 ml/day

- **For children weighing 10–20 kg:**
- Fluid requirement: **1,000 ml for the first 10 kg + 50 ml per kilogram for every kg over 10**
- Formula: Fluid needs=1000 ml+(50 ml/kg×(Weight in kg))
- Fluid needs=1000 ml+(50 ml/kg×(Weight in kg))
- **Example**: For a child weighing 15 kg:
  1000 ml+(50 ml×5 kg)=1250 ml/day

- **For children weighing more than 20 kg:**
- Fluid requirement: **1,500 ml for the first 20 kg + 20 ml per kilogram for every kg over 20**
- Formula: Fluid needs=1500 ml+(20 ml/kg×(Weight in kg))
  Example: For a child weighing 24 kg:
  1500 ml+(20 ml×4 kg)=1580 ml/day

Hourly Maintenance Fluid Calculation

- To calculate the hourly maintenance fluid requirement and fluid deficit for a child, here are the formulas broken down by weight range:
- **For the first 10 kg of body weight**:
- Requirement: **4 mL/kg per hour**
- Formula: Hourly fluid needs=Weight in kg×4 mL/kg

- **For children weighing 11–20 kg:**
- Requirement: **40 mL/hr for the first 10 kg + 2 mL/kg per hour for each kg over** 10
- **Formula:** Hourly fluid needs=40 mL+(2 mL/kg×(Weight in kg))
  Example: For a child weighing 15 kg :
  40 mL+(2 mL×5 kg)=50 mL/hr

- For children weighing above 20 kg:
- Requirement: **40 mL/hr for the first 10 kg + 20 mL/hr for the next 10 kg + 1 mL/kg per hour for each kg over 20**
- Formula:
  Hourly fluid needs=40 mL+20 mL+(1 mL/kg×(Weight in kg)
- **Example**: For a child weighing 24 kg:
  40 mL+20 mL+(1 mL×4 kg)=64 mL/hr

Daily and Hourly Maintenance Fluid Requirements

Weight	Daily Maintenance Requirement	Hourly Maintenance Requirement
1-10 kg 100 ml/kg	100 mL/kg	4 mL/kg
11-20 kg	1000 ml plus 50 ml/kg for each kg >10 kg	2 mL/kg + 2 mL/kg for each kg >10 kg
>20 kg	1500 ml plus 20 ml/kg for each kg >20 kg	1 mL/kg + 1ml/kg for each kg >20 kg

Fluid Deficit Calculation

- **The volume of fluid deficit may then be calculated using the following**
- formula:
- Calculated fluid deficit (mL) = %dehydration ×weight (kg) × 10

Fluid Deficit Calculation

% Dehydration

- This is an estimated percentage based on the **degree of dehydration** (mild, moderate, or severe).
- It is calculated using weight loss or clinical signs.
- **weight loss Example:**
  - If a child's normal weight is **12 kg**, but now weighs **11.2 kg**, the weight loss is **0.8 kg**.
  - % Dehydration = **(0.8 kg ÷ 12 kg) × 100 = 6.7% (~7%)**.

clinical signs

- **Mild Dehydration (3-5% in infants, 3-4% in children)**:

. Minimal symptoms; **normal pulse, breathing, blood pressure, and behavior**

. Slight **thirst**; **moist mucous membranes**

. **Tears and urine output present**

clinical signs

- **Moderate Dehydration (6-9% in infants, 6-8% in children)**:

. **Increased pulse, faster breathing** (tachypnea)

. **Irritable, more thirsty**

. **Dry mucous membranes, decreased tears**

. **Delayed capillary refill (2-4 sec), oliguria (low urine output)**

clinical signs

- **Severe Dehydration (≥10% in infants, 10% in children):**
- **Very increased pulse, deep rapid breathing (hyperpnea), and possible shock**
- **Hyperirritable or lethargic**
- **Absent tears, sunken eyes, sunken fontanel**
- **Very delayed capillary refill (>4 sec), cold/mottled skin**
- **Oliguria or anuria (no urine production)**

Fluid Deficit Calculation

- Calculated fluid deficit (mL) = %dehydration ×weight (kg) × 10
- e.g. **7%** dehydration in a **12 kg** child

= 7 × 12 × 10 = 840 mL deficit.

How to Use This in Clinical Practice

- **Assess the child's dehydration status** (mild, moderate, or severe).
- **Determine weight and dehydration percentage** based on clinical signs.
- **Apply the formula** to estimate fluid deficit.
- **Plan fluid replacement therapy** based on severity:

. **Mild:** Oral Rehydration Therapy (ORT)

. **Moderate/Severe:** IV fluids (e.g., normal saline or Ringer's lactate)

1. Oral Rehydration Therapy (ORT) for Mild to Moderate Dehydration

Guidelines:

- **Mild dehydration → 50 mL/kg ORS** over **4–6 hours**.
- **Moderate dehydration → 100 mL/kg ORS** over **4–6 hours**.
- **Ongoing losses (e.g., diarrhea) → 10 mL/kg ORS per stool**.

1. Oral Rehydration Therapy (ORT) for Mild to Moderate Dehydration

Key Considerations:

- Administer **small amounts (2–5 mL every 2–3 minutes)** if the child refuses or has nausea.
- **Ondansetron (Zofran)** may reduce vomiting and improve ORS tolerance.
- Use **commercial ORS solutions** (e.g., Pedialyte, WHO ORS) to **enhance sodium and water absorption**.
- For children resistant to drinking fluids, **frozen ORS (popsicles) or flavoring agents** can improve compliance.

2. Parenteral Fluid Therapy for Severe Dehydration

TABLE 22.4 Clinical Manifestations of Dehydration

Manifestation	Isotonic (Loss of Water and Salt)	Hypotonic (Loss of Salt in Excess of Water)	Hypertonic (Loss of Water in Excess of Salt)
Skin			
• Color	Gray	Gray	Gray
• Temperature	Cold	Cold	Cold or hot
• Turgor	Poor	Very poor	Fair
• Feel	Dry	Clammy	Thickened, doughy
Mucous membranes	Dry	Slightly moist	Parched
Tearing and salivation	Absent	Absent	Absent
Eyeball	Sunken	Sunken	Sunken
Fontanel	Sunken	Sunken	Sunken
Body temperature	Subnormal or elevated	Subnormal or elevated	Subnormal or elevated
Pulse	Rapid	Very rapid	Moderately rapid
Respirations	Rapid	Rapid	Rapid
Behavior	Irritable to lethargic	Lethargic or comatose; convulsions	Marked lethargy with extreme hyperirritability on stimulation

Types of IV Solutions

- **1. Crystalloids**

- **Isotonic Solutions**: Expand intravascular compartment. Examples include:
  - D5W
  - 0.9% NaCl (Normal Saline)
  - Ringer's solution
  - Lactated Ringer's solution

- **Hypertonic Solutions**: Expand the intravascular compartment more significantly and draw fluid from other areas. Examples include:
  - D10W
  - 3% NaCl
  - 5% NaCl
  - D5 0.45% NaCl
  - D5 0.9% NaCl

- **Hypotonic Solutions**: Cause fluid to shift from intravascular spaces into cells. Examples include:
  - D2.5W
  - 0.45% NaCl
  - 0.33% NaCl

- **Colloids**

. Used if blood volume doesn't improve with crystalloids. Examples include:

  - Albumin (5% and 25%)
  - Plasma protein fraction
  - Dextran

Energy/Calorie Requirements

- **Macronutrient Caloric Content**:
  - **Carbohydrates and proteins: 4 kcal/g; Fats: 9 kcal/g**

Energy/Calorie Requirements

- **Caloric Needs by Age**:
  - **Infants (0-6 months)**: 110-120 kcal/kg/day
  - **Infants (6 months - 1 year)**: ~100 kcal/kg/day
  - **Young Children (1-10 years)**: ~80 kcal/kg/day
  - **Older Children**: ~60 kcal/kg/day
  - **Adolescent Boys**: ~2,800 kcal/day
  - **Adolescent Girls**: ~2,200 kcal/day

Energy/Calorie Requirements

- **Macronutrient Caloric Content**:
  - **Carbohydrates and proteins: 4 kcal/g; Fats: 9 kcal/g**
  - Carbohydrates: Daily calories * 0.40 / 4 = grams of carbs per day.
  - Proteins: Daily calories * 0.30 / 4 = grams of proteins per day.
  - Fats: Daily calories * 0.30 / 9 = grams of fats per day

- Example Calculations:

Infants (0-6 months):example the weight of an infant is 5 kg.

- Caloric needs = 110-120 kcal/kg/day
- **Total caloric needs =** 5 kg × 110 kcal/kg = 550 kcal/day (lower end)
- **Total caloric needs =** 5 kg × 120 kcal/kg = 600 kcal/day (upper end)
- For macronutrient distribution

Carbohydrates (40%): (550×0.40)/4=55 grams

(600×0.40)/4=60 grams

Proteins (30%): (550×0.30)/4=41.25 grams

(600×0.30)/4=45 grams

Fats (30%): (550×0.30)/4=18.33 grams

(600×0.30)/4=20 grams



Milk Types and Caloric Content

Milk Type	Fat	Carbs	Protein	Calories
Whole Fat Milk	8 g	12 g	8 g	150 kcal
Low Fat Milk	5 g	12 g	8 g	120 kcal
Fat-Free Milk (Skimmed)	1 g	12 g	8 g	~90 kcal

Recommendations:

- Whole milk is suitable for children over 12 months.
- Low-fat milk (2%) can be given to children over 2 years.
- Skim milk is generally not recommended unless there's a medical need.

Breastfeeding vs. Formula Feeding

✓ **Breastfeeding**:

- **10-12 times/day** (highly digestible).
- Provides **20 kcal/oz**.

✓ **Formula Feeding**:

- **6 feeds/day**, ~2-3 oz per feed.
- **20 kcal/oz**, similar to breast milk.

📌 *Example Calculation:*
A **3-month-old, 5 kg infant** drinks **5-6 oz per feeding** (6 times/day):

- **30 oz/day × 20 kcal/oz = 600 kcal/day**
- Matches **550-600 kcal/day requirement**

Formula Adequacy for Infants

- **Quick Estimation**: Infant's age in months + 2 or 3 = ounces per feeding
  - E.g., a 3-month-old takes about 5-6 oz per feeding.
- **Caloric Content**: Both breast milk and commercial formulas contain 20 kcal/oz.

Daily Fluid Intake for Infants: Explanation and Calculation

Infants require **higher fluid intake per body weight** compared to older children and adults due to their **higher metabolic rate, greater insensible water losses, and immature kidneys**.

1. Fluid Requirements Based on Weight

- **General guideline: 75–90 mL per pound (mL/lb) per day**
- **Alternate conversion: 2.5–3 ounces per pound (oz/lb) per day**

Daily Fluid Intake for Infants: Explanation and Calculation

2. Example Calculation for a 10 kg (22 lb) Infant

Step 1: Convert Body Weight to Pounds

- **10 kg = 22 lb**

Step 2: Determine Fluid Needs in Ounces

- **Minimum:** 22 lb×2.5 oz/lb=55oz/day
- **Maximum:** 22 lb×3 oz/lb=66 oz/day
- Thus, a **10 kg (22 lb) infant needs between 55–66 oz (1,650–1,980 mL) of fluid daily**.

Daily Fluid Intake for Infants: Explanation and Calculation

3. Fluid Intake and Caloric Needs

Since **breast milk and infant formula provide both fluids and calories**, we can estimate the energy intake:

- **Average caloric content of breast milk or formula = 20 kcal per ounce**
- **Daily energy requirement = fluid intake (oz) × 20 kcal/oz**

For our example:

- **Minimum kcal:** 55×20=1,100 kcal/day
- **Maximum kcal:** 66×20=1,320 kcal/day
- Therefore, a **10 kg (22 lb) infant consumes approximately 1,100–1,320 kcal/day** through breast milk or formula.

Thank you